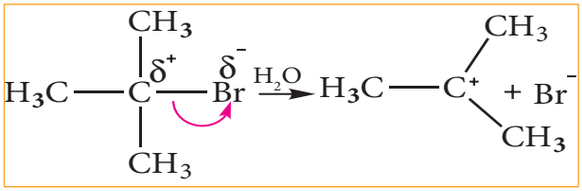
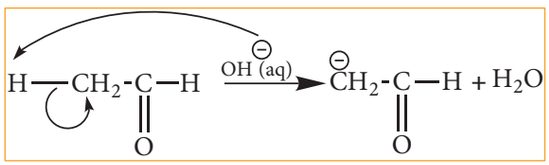
For example, in tert-butyl bromide, the C-Br bond is polar as bromine is more electronegative than carbon. The bonding electrons of the C-Br bond are attracted more by bromine than carbon. Hence, the C-Br undergoes heterolytic cleavage to form a tert-butyl cation during hydrolysis.



Let us consider the cleavage in a carbon-hydrogen (C-H) bond of aldehydes or ketones We know that the carbon is more electronegative than hydrogen and hence the heterolytic cleavage of C-H bonds results in the formation of carbanion (carbon bears a negative charge). For example in aldol condensation the OH^- ion abstracts a α-hydrogen from the aldehyde, which leads to the formation of the below mentioned carbanion.



Hybridisation of carbon in carbocation:

In a carbocation, the carbon bearing positive charge is sp^2 hybridised and hence it has a planar structure. In the reaction involving such a carbocation, the attack of a negatively charged species (nucleophiles) take place on either side of the carbocation as shown below.

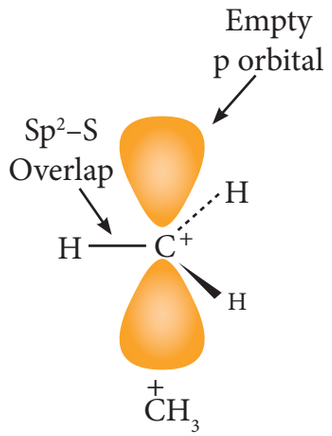


Fig 12.1(a) Shape of Carbocation

The carbanions are generally pyramidal in shape and the lone pair occupies one of the sp^3 hybridised orbitals. An alkyl free radical may be either pyramidal or planar.

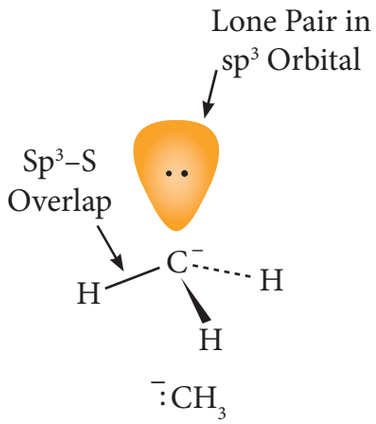


Fig 12.1(b) Shape of Carbanion

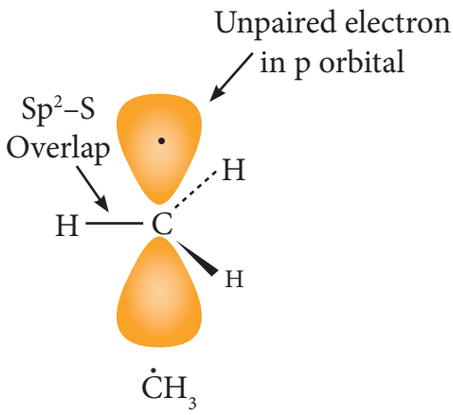


Fig 12.1(c) Shape of Carbon radical

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